

WILLIAM ORTEGA

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EDUCATION

Harvey Mudd College, Claremont, CA

Expected May 2028

Bachelor of Science, Engineering, GPA: 3.4

RELEVANT COURSEWORK: Fluid Mechanics, Statics, Experimental Engineering, Intro to Engineering Design and Manufacturing, Thermodynamics, Signals and Systems, Differential Equations and Fourier Analysis, AI-Amplified Prototyping

ENGINEERING EXPERIENCE

Undergraduate Research Assistant, Flow Imaging Lab — Harvey Mudd College

January 2026 - Present

- Demonstrated higher thrust coefficients for fully submerged horizontal fins than surface-crossing fins across most tested configurations, reaching a peak ($C_T = 4.16$), by designing a horizontal-fin test fixture and executing 100+ trials across fin thickness, amplitude, and frequency using an ATI Mini40 six-axis load cell and NI DAQ for data acquisition
- Characterized the effects of fin aspect ratio, flapping frequency, and submersion depth on thrust performance across 100+ surface-crossing trials, maintaining frequency within ± 0.02 Hz to ensure repeatable comparisons
- Redesigned the lab's MATLAB thrust-processing pipeline for horizontal configurations by remapping the thrust force axis and rewriting the inertial correction from in-air trials, enabling direct thrust-coefficient comparison against vertical cases
- Resolved trailing-edge wake vortex structures by adapting the high-speed imaging/PIV workflow for horizontal configurations, implementing spline-based reflection correction and processing velocity fields in MATLAB with PIVlab
- Presenting horizontal-fin and aspect-ratio results at the APS Division of Fluid Dynamics Annual Meeting (Nov. 2026)

Propulsion Subteam Member, Mudd Amateur Rocketry Club (MARC) — Harvey Mudd College

September 2026 - Present

- Developing CFD analysis and flight models to support propulsion design for the team's competition rocket

ENGINEERING PROJECTS

Autonomous Underwater Vehicle (AUV) – Depth, Temperature & RGB System (Team of 4)

January 2026 - May 2026

- Built and deployed a PVC-frame AUV to measure spectral light attenuation, pressure, and temperature to 3 m depth off Dana Point, CA — motivated by regional kelp-forest decline
- Designed and soldered a custom sensor/data-acquisition board integrating a TCS34725 RGB sensor (I²C), MPX5700 pressure sensor, and NTC thermistor on a Teensy 4.1, logging synchronized depth-resolved data to microSD
- Designed a two-stage MCP6002 op-amp circuit (gain ~ 6.87) scaling the pressure signal into the Teensy's 0.3–3.3 V ADC window to prevent clipping; calibrated voltage-to-depth in saltwater with a linear fit
- Implemented proportional depth control, cut motor noise via motors-off holds, and post-processed in MATLAB (Beer–Lambert, per-channel calibration) to measure $>80\%$ surface-light loss by 3 m

Mobile Lectern – Client Design Project (Team of 4)

September 2025 - November 2025

- Designed and built a lightweight, eco-friendly portable lectern to the client's needs, translated requirements into specifications and delivered the final product with a SolidWorks drawing package, BOM, and process router

Machinist's Hammer Project

September 2025 - November 2025

- Calibrated and operated manual mills and lathes to manufacture a machinist's hammer from raw steel and nylon to thousandths-of-an-inch specifications
- Applied drilling, tapping, heat treatment, and tempering processes while following technical drawings and process router

LEADERSHIP & ACTIVITIES

Founder & Chapter Co-President- SHPE Harvey Mudd College Chapter

January 2026 – Present

- Grew engagement to 52 students through recruiting, campus outreach, and professional-development programming
- Secured \$10K in institution funding to sponsor 16 members' professional development at the national SHPE conference
- Planned 3 professional-development workshops and launched company outreach to establish a sponsorship pipeline

Club President - Societal Advancement of Latinx at Harvey Mudd College

January 2025 – May 2026

- Led and planned 5+ cultural and social events each semester to foster community, belonging and appreciation
- Expanded cross-campus collaboration with five Latinx organizations to strengthen community retention and belonging

TECHNICAL SKILLS

- **Software:** SolidWorks, MATLAB, Python, FEA, LabVIEW, COMSOL, SystemVerilog, C++, OpenRocket
- **Manufacturing:** 2-axis Machining, DFM, DFA, GD&T, 3D Printing, Laser Cutting, Waterjet Cutting
- **Languages:** English (Native), Spanish (Native)